Venkat Chavan N

AI/ML Researcher | Neuro-Symbolic AI, LLM Evaluation & Multi-Agent Learning

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PROFILE

I am an AI/ML researcher with an MSc in Big Data and Artificial Intelligence from SRH University of Applied Sciences, Berlin. My work focuses on building and evaluating AI systems that move beyond surface-level pattern matching toward structured reasoning, retrieval-grounded generation, and safety-aware decision-making.

My research spans neuro-symbolic AI, LLM evaluation, hierarchical retrieval-augmented generation, multi-agent reinforcement learning for autonomous driving, Sanskrit NLP, and privacy-preserving AI systems.

RESEARCH INTERESTS

Trustworthy LLM Systems. Hallucination control, evidence grounding, RAG evaluation, citation accuracy, and adversarial robustness of retrieval-augmented language models.

Neuro-Symbolic AI & Structured Reasoning. Knowledge graphs, symbolic constraints, rule-based reasoning, and structured representations as a route to reasoning beyond surface-level pattern matching.

Multi-Agent Reinforcement Learning. Coordinated autonomous driving in mixed-traffic environments, zero-shot transfer, safety-rule evaluation, and GNN/GAT-enhanced agent coordination.

Language Model Evaluation. SanskritEval, sandhi segmentation, morphological generalisation, low-resource NLP, and benchmark design that separates linguistic structure from surface patterns.

Privacy-Preserving & Security-Aware AI. Encrypted vector operations, privacy-preserving retrieval, cyber threat intelligence, defensive AI tooling, and PII-aware evidence handling.

EDUCATION

MSc Big Data and Artificial Intelligence

SRH University of Applied Sciences, Berlin (2023 – 2025)

Thesis: Coordinated Multi-Agent Reinforcement Learning for Autonomous Vehicles in Mixed-Traffic Environments

Grade: 1.0/4.0 (S-tier)

BE Information Science and Engineering

Global Academy of Technology, Bengaluru (2018 – 2022)

PUBLICATIONS AND SCHOLARLY WORK

Encrypted Vector Operations for Privacy-Preserving Machine Learning and Data Retrieval (2025)

D. V S, S. B, M. L and V. C. Nagabhushana

2025 International Conference on Emerging Technologies in Electronics and Green Energy (ICETEG), Mysore, India. *DOI: 10.1109/ICETEG66194.2025.11473203*

Status: Published

Encrypted vector operations supporting privacy-preserving similarity search and biometric authentication.

Study on Crowd Density Estimation and Location Prediction in Public Transport System (2022)

Deepthi V S, Venkat Chavan N, Sandhya Shanbhag, Sakshi S Dandappala

International Journal of Innovative Research in Technology (IJIRT), Vol. 8, Issue 10, pp. 30–36. *PDF*

Status: Published

Survey and design study on computer-vision-based crowd density estimation for public-transport systems.

Crowd Density Estimation and Location Prediction in Public Transport System (2022)

Deepthi V S, Venkat Chavan N, Sandhya Shanbhag, Sakshi S Dandappala

International Journal of Engineering Research & Technology (IJERT), Vol. 11, Issue 7. DOI: 10.5281/zenodo.18438235. PDF

Status: Published

Implementation and evaluation of crowd density estimation and location prediction for public transport.

Coordinated Multi-Agent Reinforcement Learning for Autonomous Vehicles in Mixed-Traffic Environments (2025)

Venkat Chavan N

MSc thesis, SRH University of Applied Sciences, Berlin

Status: Submitted (Yet to be Published)

MSc thesis on CMARL with GNN/GAT-enhanced coordination and safety-rule evaluation. Defended with grade 1.0/4.0.

Hierarchical Retrieval-Augmented Generation for Cyber Threat Intelligence (2026)

Venkat Chavan N

Accepted at international conference (venue and proceedings to be added)

Status: Accepted

Hierarchical RAG system for analyst-grade cyber threat intelligence with citation-grounded reasoning. Accepted and presented at an international conference; full proceedings reference to be added.

TinyML and Edge AI (book chapter) (2026)

Venkat Chavan N

Edited volume: to be confirmed

Status: Submitted (Yet to be Published)

Book chapter on TinyML and Edge AI; details to be confirmed once contract is finalised.

RESEARCH AND PROFESSIONAL EXPERIENCE

Graduate Research Intern

SRH University of Applied Sciences, Berlin (Oct 2024 – Mar 2025)

- Research focus: Enabling AI to Understand and Process Abstract Concepts (Supervisor: Prof. Dr. Alexander Iliev).
- Investigated structured representations and symbolic reasoning for abstract-concept processing.
- Designed controlled benchmarks, holdout splits, and error taxonomies to distinguish structural learning from surface-level pattern matching.
- Developed evaluation methodology to assess whether model behaviour reflects generalisable structure or dataset shortcuts.

Master's Thesis Research: CMARL for Autonomous Driving

SRH University of Applied Sciences, Berlin (2024 – 2025)

- Designed a Coordinated Multi-Agent Reinforcement Learning framework for autonomous vehicles in mixed-traffic environments using CARLA and PyTorch.
- Investigated MADDPG-style coordination with GNN/GAT-enhanced communication and safety-rule evaluation.
- Built training, evaluation, and zero-shot generalisation pipelines with controlled traffic scenarios.
- Defended thesis with grade 1.0/4.0 (S-tier).

Data Engineering Intern

Ford Werke GmbH (Sept 2024 – Mar 2025)

- Built GCP-based ETL workflows and cloud-native pipelines in BigQuery for CRM and marketing data systems.
- Designed analyst-facing dashboards in Looker Studio for sales and marketing analytics.
- Collaborated with analytics teams on quality checks, schema validation, and reproducible data pipelines.

Data Science Intern

AI Variant (May 2023 – Aug 2023)

- Conducted exploratory data analysis and applied forecasting techniques to crude-oil price modelling.
- Developed data-visualisation interfaces using Streamlit for analytical reporting.

Graduate Trainee Engineer

Siemens (Aug 2022 – Mar 2023)

- Supported QA and conducted manual testing for industrial automation software (PCS Neo).
- Wrote structured defect logs and contributed to traceability and quality documentation.

Resource Person / Guest Speaker

Bangalore Institute of Technology, Bengaluru, India (2026)

- Delivered invited sessions for engineering students on Generative AI and Prompt Engineering.
- Contributed practical academic outreach on LLM applications, prompting workflows, and responsible use of modern AI tools.
- Received certificate of recognition for academic contribution.

RESEARCH PROJECTS

H-RAG for Cyber Threat Intelligence

RAG / Knowledge Graphs / CTI / LLM Evaluation / Hallucination Control

A hierarchical retrieval-augmented generation system for cyber threat intelligence analysis, designed to preserve document structure across executive summaries, technical details, and atomic evidence units.

Methods. FAISS IVF, Sentence-Transformers, hierarchical chunking, lightweight CTI knowledge graph, local LLM inference, citation-grounded generation.

Evaluation. Internal evaluation across factual accuracy, citation accuracy, retrieval performance, hallucination rate, and latency.

Status. Accepted at international conference, 2026

https://github.com/Venkatchavan

Coordinated Multi-Agent Reinforcement Learning for Autonomous Driving

MARL / Autonomous Driving / GNNs / Safety / Zero-shot Transfer

MSc thesis research on coordinated multi-agent reinforcement learning in mixed-traffic autonomous driving environments. Defended with grade 1.0/4.0 (S-tier).

Methods. MADDPG / MADDPG++, GAT-based coordination, CARLA mixed-traffic scenarios, safety-rule evaluators.

Evaluation. Thesis grade 1.0/4.0; internal evaluation under safety rules and zero-shot transfer scenarios.

Status. Completed: Thesis grade 1.0/4.0 (S-tier)

SanskritEval: A Benchmark for Linguistic Generalisation in Sanskrit NLP

LM Evaluation / Low-resource NLP / Morphology / Benchmark Design

Benchmark suite for evaluating language models on Sanskrit sandhi segmentation and morphological case agreement using mBERT, XLM-R, IndicBERT, and MuRIL.

Methods. Sandhi segmentation, morphological acceptability minimal pairs, transformer fine-tuning, rule-based baselines.

Evaluation. Rule-based baseline reaches 1.000 F1; transformer models evaluated under controlled splits.

Status. Completed

https://github.com/Venkatchavan/SanskritEval

Onco-Variant-Guard: Neuro-Symbolic Cancer Variant Classification

Neuro-Symbolic AI / Biomedical NLP / Knowledge Graphs / Hallucination Control

Neuro-symbolic biomedical AI system for cancer variant classification combining evidence-grounded LLM explanations, an automatically extracted rule base, and a knowledge graph.

Methods. TF-IDF baseline, automatic rule extraction, knowledge-graph reasoning, RAG, evidence-only generation.

Evaluation. Baseline classifier at 59.3% accuracy; multi-layer hallucination prevention with knowledge-graph validation.

Status. Completed

https://github.com/Venkatchavan/LLM_Healthcare

ObserveML: Observability Framework for LLM Applications

LLM Evaluation / Observability / Reliability / RAG Quality

Observability layer for LLM applications: latency, token cost, hallucination markers, RAG retrieval quality, and semantic drift, surfaced through a real-time dashboard.

Methods. ClickHouse trace storage, semantic-drift monitoring, hallucination confidence scoring, multi-language SDKs.

Evaluation. Engineering benchmarks for SDK overhead and trace-query throughput.

Status. Completed

https://github.com/Venkatchavan/ObserveML-Production-Observability

Procurement Anomaly Detection

Anomaly Detection / Explainability / Civic AI / Public Sector

Applied ML system for anomaly detection in public procurement data, combining unsupervised methods with explainability for civic transparency.

Methods. Isolation Forest, LOF, SHAP, rule-based auditing, dbt transformations.

Evaluation. Empirical evaluation on Finnish procurement data with audit-aligned thresholds.

Status. Completed

https://github.com/Venkatchavan/Procurement_Anomaly_Detection

safe-evidence-redactor: Privacy-First Redaction Library and MCP Server

Privacy / Defensive AI / PII Redaction / MCP

Defensive AI tooling for offline PII redaction in security reports, logs, and AI outputs, exposed as a library, CLI, and MCP server.

Methods. Pattern-based redaction with category allowlists, JSON path tracking, MCP integration.

Evaluation. Internal pattern-coverage benchmarks; integrated and validated by phish-triage-mcp.

Status. Released: v0.1.2

https://github.com/Venkatchavan/safe-evidence-redactor

safe-web-audit-mcp: Consent-Based Defensive Web Audit

Defensive AI / MCP / Consent-based Auditing / Security

Defensive Model Context Protocol server for consent-based, non-invasive website security audits.

Methods. MCP tooling, consent verification, SSRF guards, defensive audit checks.

Evaluation. Released open-source; consent and SSRF guards validated against test scenarios.

Status. Released: v0.2.1

https://github.com/Venkatchavan/safe-web-audit-mcp

phish-triage-mcp: Local-First Phishing Triage MCP Server

Defensive AI / MCP / Phishing Triage / Privacy

Local-first MCP server that performs static analysis of suspicious emails, SMS, and URLs without contacting external services.

Methods. Static header parsing, lookalike-domain detection, PII redaction, MCP tooling.

Evaluation. Released open-source with unit-tested detection rules.

Status. Released: v0.2.0

https://github.com/Venkatchavan/phish-triage-mcp

mcp-consent-ledger: Authorisation Proof Library for Defensive MCP Servers

Defensive AI / MCP / Consent / Authorisation

Reusable consent and authorisation-proof library for defensive MCP servers, with an auditable, local-only ledger.

Methods. SHA-256 token hashing, DNS TXT and well-known proof, SSRF-guarded fetcher.

Evaluation. Released library; verification logic validated against multi-string TXT records.

Status. Released: v0.1.1

https://github.com/Venkatchavan/mcp-consent-ledger

MV-Coach-Eval: HAR Benchmarking Platform

Evaluation / HAR / Calibration / Uncertainty

Benchmarking platform for Human Activity Recognition with calibration, robustness testing, LOSO generalisation, and Monte Carlo Dropout uncertainty.

Methods. ECE, LOSO cross-validation, Monte Carlo Dropout, robustness testing.

Evaluation. Internal benchmark coverage with Hydra-based reproducible runs.

Status. Completed

https://github.com/Venkatchavan/MV-Coach-Eval

METHODS AND TECHNICAL TOOLKIT

Machine Learning & Deep Learning. PyTorch, TensorFlow, scikit-learn, Hugging Face, MLflow.

LLMs & Retrieval. RAG, LangChain, FAISS, Sentence-Transformers, Ollama, Vector search, Prompt evaluation.

Neuro-Symbolic & Knowledge-Based AI. Knowledge graphs, Rule-based reasoning, Structured representations, Evidence-grounded explanations.

Multi-Agent Learning. MADDPG, GAT / GNNs, MARL evaluation, Autonomous driving simulation.

Data Engineering. Python, SQL, BigQuery, ETL pipelines, GCP, Looker Studio.

Research Engineering. Git, Docker, Linux, Reproducible experiments, Benchmarking, Ablation studies.

TEACHING, TALKS AND ACADEMIC OUTREACH

Resource Person / Guest Speaker: Bangalore Institute of Technology, Bengaluru, India (2026)

Invited sessions on Generative AI and Prompt Engineering for engineering students, covering LLM applications, prompting workflows, and responsible use of modern AI tools. Received a certificate of recognition for academic contribution.

HOBBY AND ENGINEERING PROJECTS

T20 Decision Intelligence Platform (IPL): Analyst-grade decision intelligence platform on IPL 2008–2025 ball-by-ball data with 12 custom metrics, calibrated ML models, and an offline RL strategy lab. [repo] [demo]

SocialMedia AD Agency: AI Ad Creation SaaS: Production-grade SaaS for automated ad creation using CrewAI multi-agent orchestration with seven specialised agents. [repo]

YouTube Chat Giveaway Bot: Desktop application for conducting giveaways from YouTube live chat with real-time monitoring and audit-friendly winner selection. [repo]

E-Commerce Data Analysis Pipeline: Cloud-native big-data pipeline on AWS EMR/S3/Athena for e-commerce analytics.

American Sign Language Detection System: CNN-based real-time ASL gesture recognition system.

Crude Oil Price Prediction: Time-series forecasting for crude oil prices using ARIMA and LSTM.

CryptoApp: Mobile Encryption Tool: Android app for text encryption using Caesar Cipher and binary encoding.

Divine Insights: NLP Analysis of the Bhagavad Gita: Sentiment analysis and RAG-based Q&A; on the Bhagavad Gita.

Enhanced Data Augmentation and Synthesis: RAG-based augmentation of review datasets using FAISS and Sentence-BERT.

Interactive Fiction Co-Author: Multi-agent collaborative storytelling tool with local AI execution.

Literary Analyst Agent: AI tool for literary analysis with semantic models and visualisations.